An LLM-Augmented Stacked Ensemble for Vietnamese Emotion Recognition on Social Media

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Abstract

Fine-grained emotion recognition from Vietnamese social-media text is a difficult natural language processing problem due to the language's low-resource status and severe class imbalance in available corpora. This paper addresses a methodological gap on this problem: existing ensembles typically fuse base-model probabilities with fixed rules or linear stacking and do not condition on the raw sentence, while large language models are used mainly as offline generators or as static prompts. We propose a two-stage framework, evaluated on UIT-VSMEC, a seven-class Vietnamese social-media emotion corpus. In Stage 1, Gemini-2.5-Flash generates and paraphrases minority-class examples (Fear, Anger, Surprise) via 10-shot prompting. In Stage 2, five heterogeneous base classifiers (PhoBERT, CafeBERT, ViBERT, Logistic Regression, LinearSVC) produce out-of-fold probability distributions, and a QLoRA-fine-tuned Gemma-2-9B-it meta-model consumes the raw sentence and those five distributions jointly. The full framework reaches 70.23% accuracy, 70.33% weighted F1, and 68.00% macro F1. On the same out-of-fold features, LogReg stacking reaches 68.43% weighted F1 and weighted voting reaches 68.73%, while the text-conditioned QLoRA Gemma meta-model reaches 70.33%.

***Index Terms*—Vietnamese NLP, emotion recognition, data augmentation, large language models, ensemble learning, social media**

I. Introduction

The rapid growth of Vietnamese social-media platforms—Facebook, TikTok, Zalo—has produced vast volumes of user-generated text carrying rich emotional signals. Mining these signals through fine-grained Emotion Recognition (ER) yields value for market research, opinion monitoring, and mental health screening, yet ER remains substantially harder than coarse sentiment polarity classification. Sentiment analysis on Vietnamese text—typically framed as a three-class task (positive, negative, neutral)—has benefited from larger annotated corpora, mature pre-trained language models, and relatively clear lexical polarity cues. Hybrid architectures such as PhoBERT combined with CNN-LSTM reach over 93% accuracy on the UIT-VSFC student-feedback corpus [1]–[3], and several deep-learning baselines report F1 scores above 90% on comparable polarity benchmarks. By contrast, fine-grained emotion recognition assigns each sentence to one of seven distinct affective states (e.g., Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, Other), where labels are linguistically subtler, more context-dependent, and frequently confounded in informal social-media writing [4]. On UIT-VSMEC, the best reported result from the original benchmark study reaches only 59.74% weighted F1 using CNN with word2vec embeddings [5]—a gap of more than 30 percentage points relative to polarity systems. Subsequent monolingual BERT models have narrowed but not closed this divide, indicating that scaling pre-training alone is insufficient for reliable fine-grained Vietnamese emotion classification.

Two interconnected obstacles drive this gap. First, Vietnamese is a low-resource language [6], [7]: annotated datasets are scarce, and existing corpora tend to be small. Second, UIT-VSMEC suffers from pronounced class imbalance: the dominant label *Enjoyment* constitutes roughly 28% of the training set, whereas minority emotions such as *Surprise* ($\sim$4.5%) and *Anger* ($\sim$6.9%) are severely underrepresented. Together, these constraints limit the robustness and generalization of individual classifiers.

These obstacles coincide with a methodological gap. Existing ensembles typically fuse class probabilities through fixed rules (voting) or linear models (logistic-regression stacking) that do not condition on the raw sentence, while large language models (LLMs) are used mainly as offline data generators or as static prompts. We address this gap with a two-stage framework. In Stage 1, Gemini-2.5-Flash [8] performs targeted data augmentation for minority emotion classes using 10-shot prompting. In Stage 2, five heterogeneous base classifiers (PhoBERT, CafeBERT, ViBERT, Logistic Regression, LinearSVC) produce out-of-fold probability estimates; a Gemma-2-9B-it meta-model is then fine-tuned with parameter-efficient QLoRA on the raw sentence and those five distributions jointly.

The main contributions of this paper are:

1) An LLM-based augmentation protocol that generates

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and paraphrases Vietnamese social-media sentences for minority emotion classes with a controlled 10-shot prompting strategy.
2) A stacked ensemble in which a QLoRA-fine-tuned Gemma meta-learner jointly consumes the raw sentence and five base-model probability distributions.
3) Empirical validation on UIT-VSMEC under a shared split and preprocessing pipeline, reaching 70.23% accuracy, 70.33% weighted F1, and 68.00% macro F1. On the same out-of-fold features, LogReg stacking reaches 68.43% weighted F1 and weighted voting reaches 68.73%. Same-protocol CafeBERT and ViSoBERT reruns and zero-shot Gemma prompting are also reported for comparison.

II. Related Work

A. Vietnamese Emotion Recognition

Ho et al. [5] introduced UIT-VSMEC, a 6,927-sentence seven-class corpus, establishing a CNN+word2vec baseline of 59.74% weighted F1. Nguyen and Nguyen [9] reached 64.40% through multi-label regression with preprocessing and key-clause extraction. Pre-trained transformers further narrowed the gap: PhoBERT [10] underpins hybrid CNN-LSTM architectures that achieve 93.24% accuracy on the polarity task [3], but the fine-grained ER on informal social-media text remains harder. CafeBERT [11] (XLM-RoBERTa trained on Vietnamese Wikipedia and news) reports 66.12% F1 on VSMEC; ViSoBERT [12] reaches 68.37% weighted F1; and a GRU–CNN–BiLSTM–LSTM ensemble [13] attains 65.79% macro F1. Despite these advances, minority emotions such as Fear, Anger, and Surprise remain difficult to classify and leave room for data-centric and ensemble improvements.

B. LLM-based Augmentation and Ensemble Methods

Ding et al. [8] survey LLM-driven data creation and coannotation, noting risks of contamination and model collapse. Ubani et al. [14] show zero-shot ChatGPT generation outperforms EDA and back-translation on SST-2 and SNIPS; Whitehouse et al. [15] extend LLM augmentation to cross-lingual settings; Üveges and Ring [16] confirm 10-shot LLM prompting yields macro-F1 gains for underrepresented emotion classes. However, prior work treats the LLM as an offline generator only, or as a static zero-/few-shot classifier, and prompt design for fine-grained Vietnamese emotion categories remains underexplored. On the ensemble side, Thin et al. [17] obtain 94.03% weighted F1 on VSFC [1] via soft-voting over five Vietnamese transformers; Niimi [18] unifies multiple LLM inferences through bagging and weighted averaging. Both fuse probabilities with fixed rules; linear logistic-regression stacking likewise does not condition on the raw sentence. Thin et al. [19] further evaluate zero- and few-shot LLM prompting for Vietnamese sentiment, while Zhang and Han [20] integrate commonsense knowledge into fine-grained aspect-level analysis, highlighting leakage risks with generative models. This leaves a methodological gap for Vietnamese emotion recognition: using an LLM both to generate minority-class data and to fuse heterogeneous ensemble probabilities while conditioning on the sentence. We address that gap with the two-stage framework in Section III.

III. Methodology

This study proposes an end-to-end pipeline for processing Vietnamese social media data and recognizing emotions in sentences, as illustrated in Fig. 1. It consists of four stages: LLM data augmentation, text preprocessing, base classifier training with cross validation stacking, and meta-model fine-tuning.

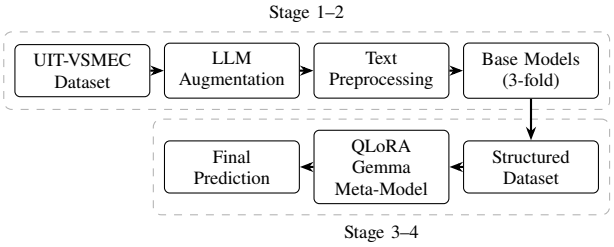


Fig. 1. Proposed pipeline: augmentation and preprocessing feed the base models; their cross validation probabilities form a structured dataset for QLoRA Gemma fine-tuning.

A. LLM-based Data Augmentation

UIT-VSMEC exhibits a 6.4:1 imbalance between the most and least frequent classes. We address this by prompting Gemini-2.5-Flash [8] via the Gemini chat interface to generate additional training examples for the three most underrepresented emotions: Fear, Anger, and Surprise. Two complementary strategies are applied: (i) *generated*—the model synthesizes entirely new Vietnamese sentences conditioned on an emotion label and 10 randomly sampled in-context examples, creating novel scenarios not present in the original corpus; (ii) *paraphrased*—the model rewrites existing annotated sentences with linguistic variation while preserving the target emotion and original semantics. Each class is augmented to a target of 1,500 examples per strategy. Augmentation operates on the raw training split before any preprocessing. Fig. 2 shows the six-stage pipeline.

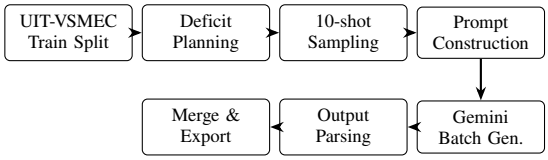


Fig. 2. Synthetic data creation pipeline for minority-class augmentation.

The pipeline operates as follows:

1) **Deficit planning.** For each minority class, the minimum number of examples required is 1,500. Other classes with numbers exceeding this target will be ignored
2) **10-shot sampling.** Ten in-class sentences are sampled uniformly at random (with replacement when the class

has fewer than ten examples) to form the few-shot context.

3) **Prompt construction.** A system message and structured human message are assembled from the selected template (generated or paraphrased).
4) **Batch generation.** The prepared prompts are sent to Gemini-2.5-Flash in groups of up to 50 at a time, so many candidate sentences can be produced in parallel rather than one by one. We set generation temperature to 0.9 to encourage varied wording across outputs [21], and apply a frequency penalty of 0.5 so the model is less likely to repeat the same phrases. Each response yields a batch of new Vietnamese sentences for the target emotion.
5) **Output parsing.** Each response is parsed line by line; only lines that follow the expected numbered format (a sequence number, a generated sentence, and a reference to the few-shot example it was based on) are kept. Lines that do not match this pattern are discarded.
6) **Merge and export.** Accepted sentences are tagged (marked as augmented with their source strategy), trimmed to the exact deficit, and concatenated with the original training split.

TABLE I
PROMPT DESIGN COMPARISON: GENERATED VS. PARAPHRASED STRATEGY

	Generated	**Paraphrased**
System role	VN social-media expert; express target emotion only	Gen-Z social-media expert; preserve meaning, tone, intent
Human task	Synthesize NEW sentences for the target emotion (in English and Vietnamese)	Rewrite sentences preserving SAME meaning and emotion
Key constraints	Natural VN style; length mix (5–10 / 10–20 / 20+ words); no other emotions	Semantic equivalence mandatory; no new facts or intensity; preserve negation and intensifiers
Output format	Numbered lines: sequence number, generated sentence, and example reference	

TABLE II
EXAMPLES OF AUGMENTED SENTENCES BY STRATEGY

Strategy	Original	Output	Emotion
Gen.	per không cho anh đi tập đâu nha mất bồ như chơi	Cái cảm giác bị bỏ rơi nó ám ảnh tui ghê lắm.	Fear
Gen.	sao mà mặt boss có thể gian phản diện đến thế nhỉ	Sao nhìn cái mặt của nhân vật phản diện nó gian manh vậy ta.	Surprise
Gen.	per bất hạnh thật tao với mày mua bánh ăn chung đi	Đời tôi khốn nạn lắm rồi, đừng ai làm phiền nữa.	Anger
Para.	sợ ngỗng còn hơn sợ chó :)))	Thà sợ chó còn hơn sợ ngỗng :)))	Fear
Para.	vl :)) còn đút tay túi áo ??	Trời đất ơi, còn đút tay vào túi áo nữa hả?	Surprise
Para.	dẹp . đéo lưu luyến .	Thôi đi. Đéo cần vương vấn gì hết.	Anger

1) Prompt Design: Both strategies share the same two-part prompt template structure: a *system message* that constrains style and emotion fidelity, and a *human message* that supplies the emotion label, its Vietnamese description, and 10 few-shot examples. Table I summarises the key differences; Table II shows representative outputs for both strategies (Gen. for Generated and Para. for Paraphrased). The shared output format requires each line to be numbered and include a reference to the few-shot example it was based on, enabling consistent parsing.

B. Text Preprocessing

Vietnamese social-media text is noisy: users mix emojis, keyboard emoticons, slang, and short forms that carry strong emotional meaning. If these signals are removed too early, models may lose useful cues. Following practices similar to prior Vietnamese social-media emotion work [6], [9], we therefore apply a shared cleaning pipeline to the training, validation, and test splits after augmentation.

Each sentence is cleaned in four steps. First, a curated dictionary of 116 emoji-to-Vietnamese mappings replaces Unicode emojis with short descriptive phrases (e.g., "vui vẻ" for happy faces and "buồn bã" for sad faces); unmatched emojis are then removed with a Unicode regex filter. Second, 39 text emoticons such as `:))`, `:(`, and `T_T` are mapped to Vietnamese words, with longer patterns matched first to avoid partial replacements. Third, 95 common internet abbreviations are expanded at the word level (e.g., *ko*/*k*/*hok*, all meaning *không*; *dc*/*đc*, meaning *được*). Fourth, URLs and HTML tags are stripped, non-Vietnamese characters are replaced with spaces while diacritics and basic punctuation are kept, repeated punctuation is collapsed, and whitespace is normalized.

After cleaning, duplicate sentences are removed so repeated content does not bias training. Stopwords are intentionally

kept. Function words such as *không* (negation) and *rất* (intensifier) often change emotion meaning, and BERT-style encoders can use these tokens through self-attention [10], [11].

C. Base Classifiers

Five heterogeneous base classifiers are trained with stratified 3-fold cross-validation, each producing probability vectors across the seven emotion classes:

- **PhoBERT-v2** [10]: pre-trained on Vietnamese Wikipedia and news; fine-tuned for 20 epochs.
- **CafeBERT** [11]: a Vietnamese BERT variant from the VLUE benchmark; best single-model accuracy on VSMEC.
- **ViBERT**: a Vietnamese BERT model further pre-trained on Vietnamese text.
- **Logistic Regression**: TF-IDF features, L2 regularization [22].
- **LinearSVC**: TF-IDF features, wrapped in a probability calibration layer to produce class probability estimates.

Each model is also refitted on the full training set to produce validation and test probability arrays. Classifier metrics are stored as JSON artifacts.

D. Stacked Ensemble with LLM Meta-Model

Weights and structured dataset. Each base model is assigned a weight proportional to its validation-set accuracy: a model's weight equals its individual accuracy divided by the sum of all five models' accuracies. This produces a set of normalized weights that sum to one, so higher-performing models contribute more to the final prediction. For each training example, a structured input is then assembled that combines the original sentence, each base model's predicted probability distribution over the seven emotion classes, and the overall weighted-average prediction. This combined representation captures both each model's individual confidence and the degree of disagreement across models. The meta-prompt therefore presents the sentence and the five probability distributions together, so the meta-learner can condition fusion on the raw text as well as on the ensemble signals.

QLoRA fine-tuning. Gemma-2-9B-it serves as the meta-learner. The model is fine-tuned with QLoRA (4-bit NF4 quantization, rank-16 adapters) using completion-only cross-entropy loss on the emotion label as the target output. Training uses the structured training dataset built from out-of-fold predictions (unbiased by construction), which mitigates information leakage into the meta-model [23]. At inference, the model generates the emotion label jointly conditioned on the sentence and the five base-model probability distributions in the structured meta-prompt.

IV. Experiments

A. Dataset and Splits

UIT-VSMEC [5] contains 6,927 human-annotated sentences drawn from Vietnamese social-media platforms, covering seven emotion labels: Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, and Other. We follow the standard 8:1:1 split protocol, yielding approximately 5,548 training, 686 validation, and 692 test sentences after preprocessing. The training set is augmented to 1,500 examples per class for the minority classes before preprocessing.

B. Baselines and Evaluation

We evaluate on the two following experimental conditions. Evaluation metrics are accuracy, weighted F1 (W-F1), and macro F1.

Augmentation ablation. CafeBERT is trained on three versions of the training data—original, paraphrased-augmented, and generated-augmented—and evaluated on the held-out test set using accuracy, weighted F1, and per-class F1 for the three minority emotions (Fear, Anger, Surprise).

Ensemble comparison. Table V has two blocks. The upper block uses our split, preprocessing, and 692-sentence test set. It compares the stacked QLoRA Gemma meta-model with (i) two conventional fusion baselines on the same five out-of-fold class-probability vectors, without the original sentence: weighted soft voting (accuracy-normalized convex combination, then $\arg\max$) and a multinomial logistic-regression stacking classifier on the concatenated 35-dimensional OOF vector, with C selected on validation macro-F1; (ii) Gemma-2-9B-it zero-shot prompting on the same structured meta-prompts; and (iii) CafeBERT and ViSoBERT rerun on the same training, validation, and test files. The lower block reports previously published UIT-VSMEC scores (CafeBERT and XLM-RoBERTa-Large from VLUE [11], ViSoBERT [12], the MLR pipeline [9], and the GRU–CNN–BiLSTM–LSTM ensemble [13]) for context only; those papers may differ in preprocessing, training, and evaluation, so we do not treat them as same-protocol baselines or as a state-of-the-art comparison. The original CNN+word2vec benchmark [5] (59.74% weighted F1) is omitted from the table for the same reason.

V. Results and Analysis

A. Impact of LLM Augmentation

Table III reports CafeBERT performance across the three training conditions. Both augmentation strategies yield consistent improvements. The generated data condition produces the largest gains, raising accuracy from 63.44% to 67.03% and weighted F1 from 62.8% to 67.1%. Minority-class gains are particularly pronounced: Anger F1 rises from 38% to 46% (a gain of 8 percentage points), confirming that synthetic generation effectively addresses the most severely imbalanced classes. Fear and Surprise also improve, though more modestly, reflecting their lower initial imbalance. Paraphrasing yields intermediate improvements, suggesting that linguistic variation alone is beneficial but less impactful than generating entirely new examples.

Table IV further analyses the quality of the generated sentences. Zero exact and near-duplicate test-set overlap confirms no evaluation leakage in the synthetic corpus. An overall fidelity score of 0.776 shows that generated text is statistically close to original social-media sentences; the KL divergence of 0.164 reflects intentional upsampling of minority classes,

not a quality failure. Lower Distinct-2/3 values reveal moderate lexical repetition from 10-shot conditioning, yet generated data still outperforms paraphrasing because novel scenarios expand minority-class coverage more than surface variation alone.

TABLE III
IMPACT OF LLM AUGMENTATION ON CAFEBERT (UIT-VSMEC TEST SET)

Training Data	Acc.	W-F1	Fear	Anger	Surprise
Original	0.6344	0.628	0.67	0.38	0.60
Paraphrased	0.6676	0.6682	0.67	0.42	0.61
Generated	**0.6703**	**0.671**	**0.68**	**0.46**	**0.61**

TABLE IV
QUALITY OF LLM-GENERATED TRAINING SENTENCES (5,010 SYNTHETIC VS. 5,548 ORIGINAL)

Metric	Original	Generated
Overall fidelity score	—	0.776
Distinct-2	0.593	0.451
Distinct-3	0.930	0.786
Mean sentence length (chars)	56.0	51.0
Train–test leakage (exact / near)	—	0 / 0
Emotion distribution KL div.	—	0.164

Fidelity = mean of stat, distribution, label, and correlation similarity.

B. Ensemble Performance

TABLE V
UIT-VSMEC TEST-SET COMPARISON

Method	Acc.	W-F1	M-F1
Same protocol (ours)			
Proposed (stacked + QLoRA Gemma)	**0.7023**	**0.7033**	**0.6800**
Weighted soft voting (ours)	0.6879	0.6873	0.6542
LogReg stacking meta (ours)	0.6864	0.6843	0.6459
Gemma zero-shot (ours)	0.6879	0.6853	0.6550
CafeBERT, same split (ours)	0.6474	0.6499	0.6142
ViSoBERT, same split (ours)	0.5997	0.5966	0.5604
Published scores (different protocol)			
MLR + preprocessing [9]	0.6436	0.6440	—
CafeBERT [11]	—	0.6612	—
ViSoBERT [12]	0.6810	0.6837	—
XLM-RoBERTa-Large [11]	0.6137	0.6020	—
GRU–CNN–BiLSTM–LSTM [13]	—	—	0.6579

Note. Upper block: same split, preprocessing, and 692-sentence test set. Lower block: previously published scores under other protocols; not used to claim state-of-the-art. W-F1: weighted F1; M-F1: macro F1; —: not reported.

Table V reports two comparisons. Under our split and preprocessing, the stacked QLoRA system reaches 70.23% accuracy, 70.33% weighted F1, and 68.00% macro F1. The fair transformer baselines are CafeBERT and ViSoBERT retrained on the same files: CafeBERT is the stronger rerun at 64.99% weighted F1, and ViSoBERT reaches 59.66%. The proposed method therefore improves by 5.34 percentage points in weighted F1 over same-protocol CafeBERT and by 10.67 points over same-protocol ViSoBERT. Published CafeBERT (66.12%) and ViSoBERT (68.37%) scores appear only in the lower block. They come from different preprocessing, training, and evaluation settings, so a 70.33% vs. 68.37% gap is not a same-protocol state-of-the-art claim; on this pipeline the ViSoBERT rerun is 59.66% weighted F1.

The two conventional fusion rows compare the QLoRA meta-learner with ordinary stacking on the same five out-of-fold class-probability vectors. Weighted soft voting applies *fixed* fusion of those vectors (already supplied to Gemma as a consensus summary). LogReg stacking trains a *linear* meta-classifier on the concatenated 35-dimensional OOF vector, without the sentence, with C selected on validation macro-F1. Zero-shot Gemma uses the same structured meta-prompts without supervised fine-tuning. QLoRA Gemma attains 70.33% weighted F1, against 68.73% for voting (+1.60) and 68.43% for LogReg (+1.90). The macro-F1 relative to LogReg is +3.41 (68.00% vs. 64.59%). All three stacking systems use the same five base models and the same OOF signals; QLoRA Gemma additionally conditions on the sentence. This comparison does not evaluate non-linear heads such as MLP or XGBoost.

A 1.80 percentage-point gain in weighted F1 over zero-shot Gemma (70.33% vs. 68.53%) shows that QLoRA fine-tuning on the structured dataset helps relative to zero-shot inference. The best same-protocol single transformer (CafeBERT, 64.99% weighted F1) trails the full ensemble by 5.34 points, showing that stacking heterogeneous BERT and TF-IDF signals goes beyond any individual classifier on this pipeline. The CafeBERT augmentation ablation in Table III (67.10% weighted F1 on generated data) is a separate training-data study and is not the same-protocol CafeBERT baseline in Table V.

VI. CONCLUSION

A. Summary

This paper proposed a two-stage framework for Vietnamese social-media emotion recognition, addressing a methodological gap in how LLMs and ensembles are combined on this task.

In Stage 1, few-shot LLM augmentation consistently improved CafeBERT's performance across all three under-represented emotion categories. Generated synthetic data—new sentences conditioned on 10 in-context examples—outperformed paraphrasing, raising weighted F1 from 62.8% to 67.1% and increasing Anger F1 by 8 percentage points (38% to 46%). A fidelity analysis confirmed that the synthetic corpus is statistically plausible (overall fidelity score 0.776, zero test-set leakage), establishing that new scenario coverage matters more than mere surface variation.

In Stage 2, stacking heterogeneous BERT and TF-IDF signals through a QLoRA-fine-tuned meta-model reached 70.23% accuracy, 70.33% weighted F1, and 68.00% macro F1 on UIT-VSMEC under our split and preprocessing. Relative to same-protocol reruns, this is 5.34 percentage points in weighted F1 above CafeBERT (64.99%) and 10.67 points above ViSoBERT (59.66%). Published transformer scores are not used as a state-of-the-art comparison because they follow different protocols. The full ensemble surpassed zero-shot Gemma prompting by

1.80 percentage points in weighted F1 (70.33% vs. 68.53%). On the same out-of-fold probability features, weighted soft voting reached 68.73% weighted F1 and logistic-regression stacking reached 68.43%. Stage 1 uses the LLM to generate minority-class data; Stage 2 uses a QLoRA meta-learner that jointly consumes the sentence and the five base-model probability distributions.

B. Future Work

Future work will explore the following directions:

1) **Cross-dataset and cross-domain generalization.** Future work should evaluate the framework on related Vietnamese NLP tasks (aspect-level sentiment, hate-speech detection, multi-label corpora) to verify that LLM augmentation and stacked ensembling generalize beyond fine-grained emotion recognition.
2) **Automated quality validation for synthetic data.** The moderate Distinct-2/3 scores observed in Table IV call for automated filtering before synthetic examples enter training: candidate filters include perplexity thresholds, emotion-consistency scoring via a held-out classifier, semantic-similarity caps relative to near-duplicates, and LLM-as-judge fluency checks. The goal is to retain minority-class coverage gains while improving lexical diversity and reducing formulaic patterns.
3) **Dynamic, uncertainty-aware ensemble weighting.** The fixed accuracy-based weighting scheme ignores per-example prediction confidence and inter-model disagreement. Replacing them with uncertainty-derived weights—computed from prediction entropy or calibrated confidence scores—may improve macro F1 on minority classes where static weights tend to favor majority-class-dominant models.
4) **Efficiency and deployability.** The current QLoRA meta-model requires substantial GPU memory at both training and inference time. Future work should investigate distilling the Gemma meta-learner into a smaller classifier or replacing it with a lightweight stacking head, making the framework practical for deployment in resource-constrained environments without sacrificing the gains from structured dataset from ensemble predictions.

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